

Guia 7 Ex 1

$$\sigma_{ij} = \lambda e_{\alpha\alpha} S_{ij} + 2\mu e_{ij}$$

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}$$

$$G = \mu = \frac{E}{2(1+\nu)}$$

$$E = \frac{\mu(3\lambda+2\mu)}{\lambda+\mu}$$

$$\nu = \frac{\lambda}{2(\lambda+\mu)}$$

Para $i \neq j$

$$\sigma_{xy} = \lambda e_{\alpha\alpha} S_{xy} + 2\mu e_{xy}$$

$$S_{xy} = 0 \text{ quando } x \neq y$$

$$\sigma_{xy} = 2\mu e_{xy}$$

$$e_{xy} = \frac{\sigma_{xy}}{2\mu} = \frac{\sigma_{xy}}{2 \cdot \frac{E}{2(1+\nu)}} = \frac{E}{2(1+\nu)} \sigma_{xy}$$

$$\left[\frac{1+\nu}{E} \sigma_{xy} = e_{xy} \right] \text{ Analogamente para } e_{yz} \text{ e } e_{zx}$$

Para $i = j$

$$\sigma_{ii} = \lambda e_{\alpha\alpha} S_{ii} + 2\mu e_{ii} \quad [e_{\alpha\alpha} = 3e_{ii} \quad S_{ii} = 1]$$

$$\sigma_{ii} = 3\lambda e_{ii} + 2\mu e_{ii}$$

$$\sigma_{ii} = e_{ii} [3\lambda + 2\mu]$$

$$e_{ii} = \frac{\sigma_{ii}}{(3\lambda + 2\mu)} \quad \textcircled{A}$$

Portanto de b. ley de hooke

$$\sigma_{xx} = \lambda e_{\alpha\alpha} S_{xx} + 2\mu e_{xx} \quad \text{Reemplazamos por } \textcircled{A} \quad S_{xx} = 1 \quad e_{\alpha\alpha} = e_{xx} + e_{yy} + e_{zz}$$

$$\sigma_{xx} = \frac{\lambda}{(3\lambda + 2\mu)} \sigma_{xx} + 2\mu e_{xx}$$

$$\sigma_{xx} = \frac{\lambda}{(3\lambda + 2\mu)} (\sigma_{xx} + \sigma_{yy} + \sigma_{zz}) + 2\mu e_{xx} \quad \text{despejamos } e_{xx}$$

$$e_{xx} = \frac{1}{2\mu} \left(\sigma_{xx} - \frac{\lambda}{(3\lambda + 2\mu)} (\sigma_{xx} + \sigma_{yy} + \sigma_{zz}) \right) \quad \text{nos faltaría común } \frac{1}{(3\lambda + 2\mu)}$$

$$e_{xx} = \frac{1}{2\mu(3\lambda + 2\mu)} \left((3\lambda + 2\mu) \sigma_{xx} - \lambda (\sigma_{xx} + \sigma_{yy} + \sigma_{zz}) \right)$$

$$e_{xx} = \frac{1}{2\mu(3\lambda+2\mu)} \cdot (2(\lambda+\mu)\sigma_{xx} - \lambda(\sigma_{yy} + \sigma_{zz}))$$

$$e_{xx} = \frac{2(\lambda+\mu)}{2\mu(3\lambda+2\mu)} \cdot \left(\sigma_{xx} - \frac{\lambda}{2(\lambda+\mu)} (\sigma_{xx} + \sigma_{zz}) \right) \quad \left[V = \frac{\lambda}{2(\lambda+\mu)} \right]$$

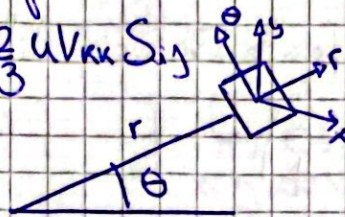
$$e_{xx} = \frac{1}{E} (\sigma_{xx} - V(\sigma_{yy} + \sigma_{zz})) \quad \text{Analogamente für } e_{yy} \text{ } e_{zz}$$

Ej 2 Guia 7

Ecuaón de Stokes para fluidos Newtonianos

$$\nabla_{ij} = -P S_{ij} + 2\mu V_{ij} - \frac{2}{3} \mu V_{kk} S_{ij}$$

Tomando coordenadas polares



$$\tau_{r\theta} = -P S_{r\theta} + 2\mu V_{r\theta} - \frac{2}{3} \mu (V_{rr} + V_{\theta\theta} + V_{zz}) S_{r\theta}$$

$$\tau_{r\theta} = 2\mu V_{r\theta} \quad (1)$$

$$V_{r\theta} = \frac{1}{2} \left(\frac{1}{r} \frac{\partial V_r}{\partial \theta} + \frac{\partial V_\theta}{\partial r} - \frac{V_\theta}{r} \right) \quad \text{con } \frac{\partial V_r}{\partial \theta} = 0$$

$$V_{r\theta} = \frac{1}{2} \frac{\partial V_\theta}{\partial r}$$

$$\frac{V_\theta}{r} \approx 0$$

porque el radio interior es pequeño en comparación con el exterior

Reemplazamos $V_{r\theta}$ en (1)

$$\tau_{r\theta} = 2\mu \frac{1}{2} \frac{\partial V_\theta}{\partial r}$$

$$\tau_{r\theta} = \mu \frac{\partial V_\theta}{\partial r} \quad (2)$$

Ecuaón de Newton para fluidos $\tau = \mu \cdot \frac{\partial V_x}{\partial y}$

Ahora analizamos la expresión para el Torque T

$$T = \int_{z=0}^{z=L} \int_{\theta=0}^{\theta=2\pi} r dF = \int_0^L \int_0^{2\pi} r \tau_{r\theta} \frac{r d\theta dz}{r} = \tau_{r\theta} r^2 \cdot 2\pi L$$

despejando $\tau_{r\theta}$ nos queda $\tau_{r\theta} = \frac{T}{2\pi L r^2}$

igualamos (2) y (3)

$$\mu \frac{\partial V_\theta}{\partial r} = \frac{T}{2\pi L r^2} \rightarrow \frac{\partial V_\theta}{\partial r} = \frac{T}{2\pi L r^2 \mu}$$

Integramos de R_i a r

$$\int_{R_i}^r \frac{\partial V_\theta}{\partial r} dr = V_\theta(r) - \underbrace{V_\theta(R_i)}_{0/\text{si es el borde}} = \int_{R_i}^r \frac{T}{2\pi r^2 L \mu} dr = \frac{T}{2\pi L \mu} \left(\frac{1}{R_i} - \frac{1}{r} \right)$$

Así obtenemos una expresión para la distribución de la velocidad

$$V_\theta(r) = \frac{T}{2\pi L \mu} \left(\frac{1}{R_i} - \frac{1}{r} \right)$$

$$V_\theta(R_e) = \omega \cdot R_e = \frac{T}{2\pi L \mu} \left(\frac{1}{R_i} - \frac{1}{R_e} \right)$$

↓
velocidad angular

Despejamos la viscosidad del fluido μ

$$\mu = \frac{T}{2\pi L \omega R_e} \left(\frac{1}{R_i} - \frac{1}{R_e} \right)$$

Despejamos T para la relación Torque velocidad angular

$$T = f(\omega) = \mu \cdot 2\pi L R_e \omega \left(\frac{1}{R_i} - \frac{1}{R_e} \right)$$

Ej 3 gwa 7

$$\sigma_x = \frac{\partial^2 \phi}{\partial y^2} \quad (1)$$

$$\sigma_y = \frac{\partial^2 \phi}{\partial x^2} \quad (2)$$

$$\tau_{xy} = -\frac{\partial^2 \phi}{\partial x \partial y} - X_y - Y_x \quad (3)$$

$$e_{xx} = \frac{1}{E} (\sigma_x - \nu \sigma_y) \quad (4)$$

$$e_{yy} = \frac{1}{E} (\sigma_y - \nu \sigma_x) \quad (5)$$

$$e_{xy} = \frac{\tau_{xy}}{G} = \frac{(1+\nu)}{E} \tau_{xy} \quad (6)$$

Las ecuaciones de compatibilidad no cumplen si y solo si

$$\nabla^4 \phi = \frac{\partial^4 \phi}{\partial x^4} + 2 \frac{\partial^4 \phi}{\partial x^2 \partial y^2} + \frac{\partial^4 \phi}{\partial y^4} = 0$$

Reemplazando (1) (2) y (3) en (4) (5) (6)

$$e_{xx} = \frac{1}{E} \left(\frac{\partial^2 \phi}{\partial y^2} - \nu \frac{\partial^2 \phi}{\partial x^2} \right)$$

$$e_{yy} = \frac{1}{E} \left(\frac{\partial^2 \phi}{\partial x^2} - \nu \frac{\partial^2 \phi}{\partial y^2} \right)$$

$$e_{xy} = \frac{(1+\nu)}{E} \left(-\frac{\partial^2 \phi}{\partial x \partial y} - X_y - Y_x \right)$$

Ecuaciones de compatibilidad

$$\frac{\partial^2 e_{xx}}{\partial y^2} + \frac{\partial^2 e_{yy}}{\partial x^2} - 2 \frac{\partial^2 e_{xy}}{\partial x \partial y} = 0 \quad \text{Reemplazando } e_{xx} \ e_{yy} \ e_{xy}$$

$$\frac{1}{E} \left(\frac{\partial^4 \phi}{\partial y^4} - \nu \frac{\partial^4 \phi}{\partial x^2 \partial y^2} \right) + \frac{1}{E} \left(\frac{\partial^4 \phi}{\partial x^4} - \nu \frac{\partial^4 \phi}{\partial x^2 \partial y^2} \right) + 2 \frac{(1+\nu)}{E} \left(\frac{\partial^4 \phi}{\partial x^2 \partial y^2} \right) = 0$$

$$\frac{\partial^4 \phi}{\partial y^4} + \frac{\partial^4 \phi}{\partial x^4} - 2\nu \frac{\partial^4 \phi}{\partial x^2 \partial y^2} + 2 \frac{\partial^4 \phi}{\partial x^2 \partial y^2} + 2\nu \frac{\partial^4 \phi}{\partial x^2 \partial y^2} = 0$$

$$\frac{\partial^4 \phi}{\partial y^4} + \frac{\partial^4 \phi}{\partial x^4} + 2 \frac{\partial^4 \phi}{\partial x^2 \partial y^2} = 0 = \nabla^4 \phi$$

b)

$$\phi_1(x, y) = ax^2 + bxy + cy^2$$

$$\frac{\partial \phi_1}{\partial x} = 2ax + by$$

$$\frac{\partial \phi_1}{\partial y} = bx + 2cy$$

$$\frac{\partial^2 \phi_1}{\partial x^2} = 2a$$

$$\frac{\partial^2 \phi_1}{\partial y^2} = 2c$$

$$\frac{\partial^3 \phi_1}{\partial x^3} = 0$$

$$\frac{\partial^3 \phi_1}{\partial y^3} = 0$$

$$\nabla^4 \phi_1 = \frac{\partial^4 \phi}{\partial y^4} + \frac{\partial^4 \phi}{\partial x^4} + 2 \frac{\partial^4 \phi}{\partial x^2 \partial y^2} = 0 + 0 + 2 \cdot 0 = 0 \quad \checkmark \text{ as de Airy}$$

$$\phi_2 = ax^3 + bx^2y + cxy^2 + dy^3$$

$$\frac{\partial \phi_2}{\partial x} = 3ax^2 + 2bxy + cy^2$$

$$\frac{\partial \phi_2}{\partial y} = bx^2 + 2cxy + 3y^2d$$

$$\frac{\partial^2 \phi_2}{\partial x^2} = 6ax + 2by$$

$$\frac{\partial^2 \phi_2}{\partial y^2} = 2cx + 6dy$$

$$\frac{\partial^3 \phi_2}{\partial x^3} = 6a$$

$$\frac{\partial^3 \phi_2}{\partial y^3} = 6d$$

$$\frac{\partial^4 \phi_2}{\partial x^4} = 0$$

$$\frac{\partial^4 \phi_2}{\partial y^4} = 0$$

$$\nabla^4 \phi_2 = \frac{\partial^4 \phi}{\partial y^4} + \frac{\partial^4 \phi}{\partial x^4} + 2 \frac{\partial^4 \phi}{\partial x^2 \partial y^2} = 0 + 0 + 2 \cdot 0 = 0 \quad \checkmark \text{ as de Airy}$$

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c) $\phi(x,y) = ax^4 + bx^3y + cx^2y^2 + dxy^3 + ey^4$

$$\frac{\partial \phi}{\partial x} = 4ax^3 + 3bx^2y + 2cxy^2 + dy^3$$

$$\frac{\partial^2 \phi}{\partial x^2} = 12ax^2 + 6bxy + 2cy^2$$

$$\frac{\partial^3 \phi}{\partial x^3} = 24ax + 6by$$

$$\frac{\partial^4 \phi}{\partial x^4} = 24a$$

$$\frac{\partial \phi}{\partial y} = bx^3 + 2cx^2y + 3dxy^2 + 4ey^3$$

$$\frac{\partial^2 \phi}{\partial y^2} = 2cx^2 + 6dxy + 12ey^2$$

$$\frac{\partial^3 \phi}{\partial y^3} = 6dx + 24ey$$

$$\frac{\partial^4 \phi}{\partial y^4} = 24e$$

$$\frac{\partial^3 \phi}{\partial x^2 \partial y} = 6bx + 4cy \rightarrow \frac{\partial^4 \phi}{\partial x^2 \partial y^2} = 4c$$

$$\nabla^4 \phi = \frac{\partial^4 \phi}{\partial y^4} + \frac{\partial^4 \phi}{\partial x^4} + 2 \cdot \frac{\partial^4 \phi}{\partial x^2 \partial y^2} = 0$$

$$24e + 24a + 2 \cdot 4c = 0$$

$$24e + 24a + 8c = 0$$

$$[3e + 3a + c = 0] \rightarrow \text{Condición para que sea de Airy}$$

$$d) \phi(x,y) = ax^5 + bx^4y + cx^3y^2 + dx^2y^3 + exy^4 + fy^5$$

$$\frac{\partial \phi}{\partial x} = 5ax^4 + 4bx^3y + 3cx^2y^2 + 2dxy^3 + ey^4$$

$$\frac{\partial \phi}{\partial y} = bx^4 + 2cx^3y + 3dx^2y^2 + 4exy^3 + 5fy^4$$

$$\frac{\partial^2 \phi}{\partial x^2} = 20ax^3 + 12bx^2y + 6cxy^2 + 2dy^3$$

$$\frac{\partial^2 \phi}{\partial y^2} = 2cx^3 + 6dx^2y + 12exy^2 + 20fy^3$$

$$\frac{\partial^3 \phi}{\partial x^3} = 60ax^2 + 24bxy + 6cy^2$$

$$\frac{\partial^3 \phi}{\partial y^3} = 6dx^2 + 24exy + 60fy^2$$

$$\frac{\partial^4 \phi}{\partial x^4} = 120ax + 24by$$

$$\frac{\partial^4 \phi}{\partial y^4} = 24ex + 120fy$$

$$\frac{\partial^3 \phi}{\partial x^2 \partial y} = 12bx^2 + 12cxy + 6dy^2$$

$$\frac{\partial^4 \phi}{\partial x^2 \partial y^2} = 12cx + 12dy$$

$$\nabla^4 \phi = \frac{\partial^4 \phi}{\partial x^4} + \frac{\partial^4 \phi}{\partial y^4} + 2 \frac{\partial^4 \phi}{\partial x^2 \partial y^2} = 0$$

$$120ax + 24by + 24ex + 120fy + 2(12cx + 12dy) = 0$$

$$120ax + 24by + 24ex + 120fy + 24cx + 24dy = 0$$

$$[(120a + 24e + 24c)X + (24b + 120f + 24d)Y] = 0$$

condición para que sea de Airy

Guia 7 Exercício 4

$$\phi = \frac{a_2}{2} x^2 + \frac{b_2}{2} xy + \frac{c_2}{2} y^2$$

$$\frac{\partial \phi}{\partial x} = a_2 x + \frac{b_2}{2} y$$

$$\frac{\partial \phi}{\partial y} = \frac{b_2}{2} x + c_2 y$$

$$\frac{\partial^2 \phi}{\partial x^2} = a_2$$

$$\frac{\partial^2 \phi}{\partial y^2} = c_2$$

$$\frac{\partial^3 \phi}{\partial x^3} = 0$$

$$\frac{\partial^3 \phi}{\partial y^3} = 0$$

$$\frac{\partial^3 \phi}{\partial x^2 \partial y} = 0 \rightarrow \frac{\partial^4 \phi}{\partial x^2 \partial y^2} = 0$$

$$\nabla^4 \phi = \frac{\partial^4 \phi}{\partial x^4} + \frac{\partial^4 \phi}{\partial y^4} + 2 \frac{\partial^4 \phi}{\partial x^2 \partial y^2} = 0$$

$$0 + 0 + 2 \cdot 0 = 0 \quad \checkmark$$

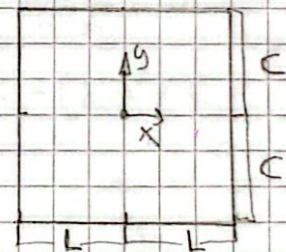
Esse é Airy

Condições de borda para $x = \pm L$ $y = \pm C$

$$\sigma_x = \frac{\partial^2 \phi}{\partial y^2} \quad \sigma_y = \frac{\partial^2 \phi}{\partial x^2} \quad \tau_{xy} = -\frac{\partial^2 \phi}{\partial x \partial y} = -\frac{1}{2} \left(\frac{\partial \sigma_x}{\partial y} + \frac{\partial \sigma_y}{\partial x} \right)$$

$$\sigma_x = c_2 \quad \sigma_y = a_2 \quad \tau_{xy} = -b_2$$

$$\sigma = \begin{bmatrix} c_2 & -b_2 \\ -b_2 & a_2 \end{bmatrix} \quad \text{Aplicamos as condições}$$

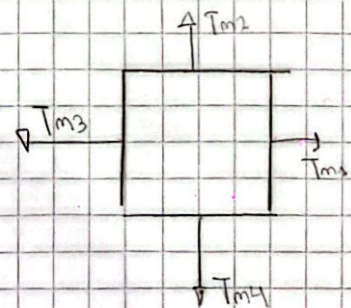


$$V_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad T_1 = \begin{bmatrix} c_2 & -b_2 \\ -b_2 & a_2 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} c_2 \\ -b_2 \end{pmatrix}$$

$$V_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad T_2 = \begin{bmatrix} c_2 & -b_2 \\ -b_2 & a_2 \end{bmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -b_2 \\ a_2 \end{pmatrix}$$

$$V_3 = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \rightarrow T_3 = \begin{pmatrix} -c_2 \\ -b_2 \end{pmatrix}$$

$$V_4 = \begin{pmatrix} 0 \\ -1 \end{pmatrix} \rightarrow T_4 = \begin{pmatrix} b_2 \\ -a_2 \end{pmatrix}$$



Para C

$$\text{para } T_{m2} \begin{pmatrix} C_2 \\ -b_2 \end{pmatrix} P = \begin{pmatrix} T_{m2} \int_{-c}^c C_2 dy \\ -T_{m2} \int_{-c}^c b_2 dy \end{pmatrix} \rightarrow \begin{pmatrix} 2 T_{m2} C_2 \\ -2 T_{m2} C b_2 \end{pmatrix}$$

$$\text{para } T_{m3} \begin{pmatrix} -C_2 \\ b_2 \end{pmatrix} P = \begin{pmatrix} -T_{m3} \int_{-c}^c C_2 dy \\ T_{m3} \int_{-c}^c b_2 dy \end{pmatrix} \rightarrow \begin{pmatrix} -2 T_{m3} C_2 \\ 2 T_{m3} C b_2 \end{pmatrix}$$

Para L

$$\text{para } T_{m2} \begin{pmatrix} -b_2 \\ a_2 \end{pmatrix} P = \begin{pmatrix} -T_{m2} \int_{-L}^L b_2 dx \\ T_{m2} \int_{-L}^L a_2 dx \end{pmatrix} = \begin{pmatrix} -2 T_{m2} L b_2 \\ 2 T_{m2} L a_2 \end{pmatrix}$$

$$\text{para } T_{m4} \begin{pmatrix} b_2 \\ -a_2 \end{pmatrix} P = \begin{pmatrix} T_{m4} \int_{-L}^L b_2 dx \\ -T_{m4} \int_{-L}^L a_2 dx \end{pmatrix} = \begin{pmatrix} 2 T_{m4} L b_2 \\ -2 T_{m4} L a_2 \end{pmatrix}$$

Gula 7 § 5

$$\phi = \frac{a_3}{6} X^3 + \frac{b_3}{2} X^2 Y + \frac{c_3}{2} X Y^2 + \frac{d_3}{6} Y^3$$

$$\frac{\partial \phi}{\partial X} = \frac{1}{2} a_3 X^2 + b_3 X Y + \frac{c_3}{2} Y^2$$

$$\frac{\partial \phi}{\partial Y} = \frac{b_3}{2} X^2 + c_3 X Y + \frac{1}{2} d_3 Y^2$$

$$\frac{\partial^2 \phi}{\partial X^2} = a_3 X + b_3 Y$$

$$\frac{\partial^2 \phi}{\partial Y^2} = c_3 X + d_3 Y$$

$$\frac{\partial^3 \phi}{\partial X^3} = \frac{2a_3}{3}$$

$$\frac{\partial^3 \phi}{\partial Y^3} = \frac{2d_3}{3}$$

$$\frac{\partial^4 \phi}{\partial X^4} = 0$$

$$\frac{\partial^4 \phi}{\partial Y^4} = 0$$

$$\frac{\partial^3 \phi}{\partial X^2 \partial Y} = b_3 \quad \frac{\partial^4 \phi}{\partial X^2 \partial Y^2} = 0$$

$$\nabla^4 \phi = \frac{\partial^4 \phi}{\partial Y^4} + \frac{\partial^4 \phi}{\partial X^4} + 2 \frac{\partial^4 \phi}{\partial X^2 \partial Y^2} = 0 + 0 + 2 \cdot 0 = 0 \quad \checkmark \text{ Eq. 1 satisfied}$$

$$\sigma_x = \frac{\partial^2 \phi}{\partial Y^2}$$

$$\sigma_y = \frac{\partial^2 \phi}{\partial X^2}$$

$$\tau_{xy} = -\frac{\partial^2 \phi}{\partial X \partial Y} - \frac{1}{2} \frac{\partial^2 \phi}{\partial X^2} - \frac{1}{2} \frac{\partial^2 \phi}{\partial Y^2}$$

$$\sigma_x = c_3 X + d_3 Y$$

$$\sigma_y = \frac{2}{3} a_3 X + b_3 Y$$

$$\tau_{xy} = -(b_3 X + c_3 Y)$$

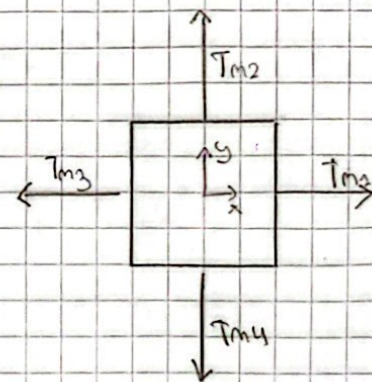
$$\sigma = \begin{bmatrix} c_3 X + d_3 Y & -(b_3 X + c_3 Y) \\ -(b_3 X + c_3 Y) & \frac{2}{3} a_3 X + b_3 Y \end{bmatrix}$$

$$V_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow T_{m2} = \sigma \cdot V_1 = \begin{bmatrix} c_3 X + d_3 Y \\ -(b_3 X + c_3 Y) \end{bmatrix}$$

$$V_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow T_{m2} = \sigma \cdot V_2 = \begin{bmatrix} -(b_3 X + c_3 Y) \\ \frac{2}{3} a_3 X + b_3 Y \end{bmatrix}$$

$$V_3 \begin{pmatrix} -1 \\ 0 \end{pmatrix} \rightarrow T_{m3} = \sigma \cdot V_3 = \begin{bmatrix} -(c_3 X + d_3 Y) \\ (b_3 X + c_3 Y) \end{bmatrix}$$

$$V_4 \begin{pmatrix} 0 \\ -1 \end{pmatrix} \rightarrow T_{m4} = \sigma \cdot V_4 = \begin{bmatrix} (b_3 X + c_3 Y) \\ -(\frac{2}{3} a_3 X + b_3 Y) \end{bmatrix}$$



Para C

$$\text{Para } T_{m1} \quad P = \begin{pmatrix} T_{m1} \int_{-c}^c (C_3 X + d_3 Y) dy \\ T_{m1} \int_{-c}^c (b_3 X + C_3 Y) dy \end{pmatrix} = \begin{pmatrix} T_{m1} C_3 X + d_3 2CT_{m1} \\ -T_{m1} b_3 X + T_{m1} C_3 2C \end{pmatrix}$$

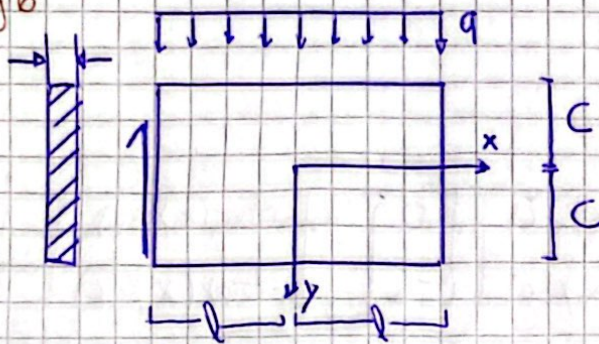
$$\text{Para } T_{m3} \quad P = \begin{pmatrix} T_{m3} \int_{-c}^c (-C_3 X + d_3 Y) dy \\ T_{m3} \int_{-c}^c (b_3 X + C_3 Y) dy \end{pmatrix} = \begin{pmatrix} -2C_3 C T_{m3} X \\ + 2b_3 C X T_{m3} \end{pmatrix}$$

Para L

$$\text{Para } T_{m2} \quad P = \begin{pmatrix} T_{m2} \int_{-L}^L -(b_3 X + C_3 Y) dx \\ T_{m2} \int_{-L}^L (-d_3 X + b_3 Y) dx \end{pmatrix} = \begin{pmatrix} -2T_{m2} C_3 LY \\ 2T_{m2} b_3 LY \end{pmatrix}$$

$$\text{Para } T_{m4} \quad P = \begin{pmatrix} T_{m4} \int_{-L}^L (b_3 X + C_3 Y) dx \\ T_{m4} \int_{-L}^L (-d_3 X + b_3 Y) dx \end{pmatrix} = \begin{pmatrix} 2T_{m4} C_3 LY \\ -2T_{m4} b_3 LY \end{pmatrix}$$

Gua 7 E6



Para determinar si la solución es correcta se debe

- 1) Verificar el equilibrio
- 2) Verificar los cond de borde
- 3) Verificar la ecuación de compatibilidad de deformaciones (esta no se pide)

$$1) \frac{\partial \Pi}{\partial X_i} + X_i^0 = 0$$

$$\frac{\partial \Pi_{xx}}{\partial x} + \frac{\partial \Pi_{xy}}{\partial y} = 0 \quad (1) \quad \left. \begin{array}{l} \frac{\partial \Pi_{xx}}{\partial x} = -\frac{2qxy}{2I} = -\frac{q}{I}xy \\ \frac{\partial \Pi_{xy}}{\partial y} = \frac{q}{I}xy \end{array} \right\}$$

$$\frac{\partial \Pi_{yy}}{\partial y} + \frac{\partial \Pi_{xy}}{\partial x} = 0 \quad (2) \quad \left. \begin{array}{l} \frac{\partial \Pi_{yy}}{\partial y} = \frac{q}{2I}(c^2 - y^2) \\ \frac{\partial \Pi_{xy}}{\partial x} = -\frac{q}{2I}(c^2 - y^2) \end{array} \right\}$$

$$\frac{\partial \Pi_{yy}}{\partial y} = \frac{q}{2I}(c^2 - y^2)$$

$$\frac{\partial \Pi_{xy}}{\partial x} = -\frac{q}{2I}(c^2 - y^2)$$

$$(1) \quad -\frac{q}{I}xy + \frac{q}{I}xy = 0$$

$$0 = 0 \quad \checkmark$$

$$(2) \quad \frac{q}{2I}(c^2 - y^2) - \frac{q}{2I}(c^2 - y^2) = 0$$

$$0 = 0 \quad \checkmark$$

Se Verifica el equilibrio

2) Verificação do bordo

bordo do Anelito ($Y = -c$)

$$T = \begin{bmatrix} 0 \\ +q \end{bmatrix} \quad M = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$T = \sigma_{xx}(X, Y=-c) \quad \sigma_{xy}(X, Y=-c) \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -\sigma_{xy}(X, Y=-c) \\ -\sigma_{yy}(X, Y=-c) \end{bmatrix} = \begin{bmatrix} 0 \\ +q \end{bmatrix}$$

Verificação

$$-\sigma_{xy}(X, Y=-c) = -\left(-\frac{q}{2I} \cdot (c^2 - (-c)^2)\right) \cdot X$$

$$-\sigma_{xy}(X, Y=-c) = -\left(-\frac{q}{2I} \cdot \underbrace{(c^2 - c^2)}_0\right) \cdot X$$

$$-\sigma_{xy}(X, Y=-c) = 0 \quad \checkmark$$

$$-\sigma_{yy}(X, Y=-c) = -\left(-\frac{q}{2I} \cdot \left(\frac{1}{3}(-c)^3 - c^2(-c) + \frac{2}{3}c^3\right)\right)$$

$$= -\left(-\frac{q}{2I} \cdot \left(-\frac{1}{3}c^3 + c^3 + \frac{2}{3}c^3\right)\right)$$

$$= -\left(-\frac{q}{\cancel{\frac{4}{3}c^3}} \cdot \left(\frac{4}{3}c^3\right)\right)$$

$$-\sigma_{yy}(X, Y=-c) = q \quad \checkmark$$

Borde do Abajo $Y=c$

$$T = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad m = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$T = n \cdot m = \begin{bmatrix} \sigma_{xy}(X, Y=c) \\ \sigma_{yx}(X, Y=c) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\sigma_{xy}(X, Y=c) = \frac{-q}{2L} \underbrace{(C^2 - C^2)}_0 \cdot X$$

$$\sigma_{xy}(X, Y=c) = 0 \quad \checkmark$$

$$\sigma_{yy}(X, Y=c) = \frac{-q}{2L} \left(\frac{1}{3} C^3 - C \cdot C + \frac{2}{3} C^3 \right)$$

$$\sigma_{yy}(X, Y=c) = \frac{-q}{2L} \left(\frac{4}{3} C^3 - C^2 \right)$$

$$\sigma_{yy}(X, Y=c) = 0 \quad \checkmark$$

Borde da direita $(X=L)$

$$T = \begin{bmatrix} 0 \\ V \end{bmatrix}; \quad m = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{onde } V \text{ é um valor diferente}$$

$$T = n \cdot m = \begin{bmatrix} \sigma_{xx}(X=L, Y) \\ \sigma_{xy}(X=L, Y) \end{bmatrix} = \begin{bmatrix} 0 \\ V \end{bmatrix}$$

$$\sigma_{xx}(X=L, Y) = \frac{q}{2L} \underbrace{(L^2 - L^2)}_0 \cdot Y + \frac{q}{2L} \left(\frac{2}{3} Y^3 - \frac{2}{5} C^2 Y \right)$$

$$\sigma_{xx}(X=L, Y) = \frac{q}{2L} \left(\frac{2}{3} Y^3 - \frac{2}{5} C^2 Y \right) \neq 0$$

Não resolve totalmente. Se pode analisar se a resultante do força e momento de cruz são zero.

lado derecho

$$T(X=L, Y) = \begin{bmatrix} \rho_x(X=L, Y) \\ \rho_{xy}(X=L, Y) \end{bmatrix}$$

normal
shear

$$\begin{bmatrix} N \\ Q \end{bmatrix} = \iint_A T(X=L, Y) dA = \int_Y \int_Z T(X=L, Y) dz$$

normal
shear

$$\begin{bmatrix} N \\ Q \end{bmatrix} = \int_{-c}^c \int_0^1 1 dz dy = \int_{-c}^c T(X=L, Y) dy$$

$$\begin{bmatrix} N \\ Q \end{bmatrix} = \begin{bmatrix} \int_{-c}^c T_x(X=L, Y) dy \\ \int_{-c}^c T_y(X=L, Y) dy \end{bmatrix} = \begin{bmatrix} \int_{-c}^c \rho_x(X=L, Y) dy \\ \int_{-c}^c \rho_{xy}(X=L, Y) dy \end{bmatrix} \rightarrow \begin{matrix} LON \\ LDQ \end{matrix}$$

LON (lado derecho normal)

$$\int_{-c}^c \rho_x(X=L, Y) dy = \int_{-c}^c \frac{q}{2I} \left(\frac{2}{3} Y^3 - \frac{2}{5} C^2 Y \right) dy = \frac{q}{2I} \left(\frac{1}{6} Y^4 - \frac{1}{5} C^2 Y^2 \right) \Big|_{-c}^c = 0$$

LDQ (lado derecho corte)

$$\int_{-c}^c \rho_{xy}(X=L, Y) dy = \int_{-c}^c \frac{-qL}{2I} (C^2 - Y^2) dy = \frac{-qL}{2I} \left(C^2 Y - \frac{Y^3}{3} \right) \Big|_{-c}^c = -qL$$

Fuerza del lado derecho

$$F_0 = \begin{bmatrix} N \\ Q \end{bmatrix} = \begin{bmatrix} 0 \\ -qL \end{bmatrix} \rightarrow \text{compara con el sentido de la fuerza dada en el problema}$$

Resultado de momento del lado derecho

$$M = \iint_A T_x dA \cdot Y = \iint_A T_x Y dA$$

$$= \int_{-c}^c \rho_x(X=L, Y) Y dy = \int_{-c}^c \frac{q}{2I} \left(\frac{2}{3} Y^4 - \frac{2}{5} C^2 Y^2 \right) dy = \frac{q}{2I} \left(\frac{2}{3} \frac{Y^5}{5} - \frac{2}{5} C^2 \frac{Y^3}{3} \right) \Big|_{-c}^c$$

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$$= \frac{q}{2l} \left[\left(\frac{2}{3} \frac{C^5}{5} - \frac{2}{5} \frac{C^5}{3} \right) - \left(\left(-\frac{2}{3} \frac{C^5}{5} \right) + \frac{2}{5} \frac{C^5}{3} \right) \right]$$

$$= \frac{q}{2l} \cdot \left[\frac{2}{15} C^5 - \frac{2}{15} C^5 + \frac{2}{15} C^5 - \frac{2}{15} C^5 \right]$$

$$= 0$$

quedan resfuerzas los cord de borde para el lado derecho

lado izquierdo

$$T = \begin{bmatrix} 0 \\ V \end{bmatrix} \quad M = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$T = N \cdot m = \begin{bmatrix} -\sigma_{xx}(X=L, Y) \\ -\sigma_{xy}(X=L, Y) \end{bmatrix} = \begin{bmatrix} 0 \\ V \end{bmatrix}$$

$$\begin{bmatrix} N \\ Q \end{bmatrix} = \iint_A T(X=L, Y) dA = - \begin{bmatrix} \int_{-c}^c N_x(X=L, Y) dy \\ \int_{-c}^c N_{xy}(X=L, Y) dy \end{bmatrix} = \begin{bmatrix} 0 \\ \int_{-c}^c N_{xy}(X=L, Y) dy \end{bmatrix}$$

$$\begin{bmatrix} N \\ Q \end{bmatrix} = \begin{bmatrix} 0 \\ -qL \end{bmatrix} \text{ coincide con distribuci3n de la fuerza dada en el problema}$$

Resultado de momento lado izquierdo

$$M = \iint_A T_x(X=L, Y) \cdot Y dA = \int_{-c}^c -N_x(X=L, Y) \cdot Y dy = - \left[\int_{-c}^c N_x(X=L, Y) Y dy \right]$$

$$M = -(0) = 0$$

quedan resfuerzas los cord de borde para el lado izquierdo

Se puede concluir que la reducci3n planteada en el ejercicio es correcta